Gabriel Kane

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OBJECTIVE

Shifting from teaching into data science and analyst work, using quantitative skills, a knack for distilling data and knowledge into key insights, and a joy for tough challenges.

EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge, MA
Bachelor of Science (B.S.) Mathematics and Music (Double Major), 2021

- ***GPA:*** *4.8 out of 5.0*
- ***Coursework:*** *Probability and Random Variables, Linear Algebra, Real Analysis, Differential Equations, Complex Analysis, and programming in Python.*

EXPERIENCE

Maternal Health Analytics, Well on Their Way, 2026

- *Turned fragmented Excel health records for a maternal-health nonprofit in Gulu, Uganda into a Python pipeline, cleaned QGIS boundaries, and Tableau dashboards for training coverage and district impact.*

Encore, 2025

- *Shipped features and fixes on a pre-launch startup codebase in SQL, React, HTML, and JavaScript, and took part in design and feature discussions.*

Private Instructor, 2024 - 2026

- *Tutored calculus (AP and college-level) and statistics.*

Mathematics Faculty, Cambridge School of Weston, Weston, MA, 2021 - 2024

- *Taught high-school mathematics, and wrote problem sets, assessments, and visualizations in LaTeX and Python.*

PROJECTS

Elementary Math Worksheet Generator, 2026,
github.com/gkane1234/elementary_math

- *Built a web app that generates printable math worksheets with randomized questions, KaTeX preview, and PDF export, from grade 6 through calculus.*

Compute Shader Barnes–Hut N-Body, 2025 - 2026,
github.com/gkane1234/gravity

- *Real-time Barnes–Hut n-body simulation with compute shaders, built to handle tens of millions of bodies across a Java, GLSL, and Python codebase.*

Inequivalent Expression Solver, 2024 - 2026,
github.com/gkane1234/Inequivalent

- *Builds algebraically inequivalent expressions and searches for ways to hit a goal with +, −, ×, and ÷. Custom structures and compression brought a saved file from about 1.7 GB to about 100 MB, and load time from about 20 seconds to about 100 ms.*

SKILLS

Python, Pandas, SQL, Java, JavaScript, React, GLSL, Tableau, QGIS, GeoPandas, Git, LaTeX, Excel, Data Cleaning, Data Transformation